

Groups and Lie Algebras: Solutions to Exercise Sheet 11

Exercise 1. Highest weight modules are unique

Solution. Let $\mathfrak{g}$ be a complex simple Lie algebra, and fix a Cartan subalgebra $\mathfrak{h}$ and a choice of positive roots. Let V and W be irreducible finite-dimensional highest weight modules with the same highest weight λ .

The statement we want to prove is

$$V \cong W.$$

The idea is exactly the same as for $\mathfrak{sl}(3)$: a highest weight vector generates the whole irreducible module, and all relations among descendants are forced by the highest weight.

Let $v_\lambda \in V$ and $w_\lambda \in W$ be highest weight vectors. Thus

$$hv_\lambda = \lambda(h)v_\lambda, \quad hw_\lambda = \lambda(h)w_\lambda \quad (h \in \mathfrak{h}),$$

and

$$\mathfrak{g}_\alpha v_\lambda = 0, \quad \mathfrak{g}_\alpha w_\lambda = 0 \quad (\alpha > 0).$$

Here $\mathfrak{g}_\alpha$ denotes the root space of root α .

By the triangular decomposition,

$$\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+,$$

where $\mathfrak{n}_+$ is the sum of positive root spaces and $\mathfrak{n}_-$ is the sum of negative root spaces. Since v_λ is killed by $\mathfrak{n}_+$ and is an eigenvector for $\mathfrak{h}$, every vector generated from v_λ is obtained by applying lowering operators from $\mathfrak{n}_-$.

Since the negative root spaces are generated by the simple negative root operators, every vector in a highest weight module is a linear combination of vectors of the form

$$F_{\alpha_{i_1}} F_{\alpha_{i_2}} \cdots F_{\alpha_{i_k}} v_\lambda,$$

where $F_{\alpha_i} = E^{-\alpha_i}$ are the lowering operators corresponding to the simple roots. For example, in the A_2 case one obtains the remaining negative root vector from

$$[E^{-\alpha_1}, E^{-\alpha_2}] = E^{-(\alpha_1 + \alpha_2)}.$$

This is exactly the construction used in the lecture notes for $\mathfrak{sl}(3)$.

We therefore define a linear map

$$\Phi : V \rightarrow W$$

by

$$\Phi(v_\lambda) = w_\lambda$$

and, more generally,

$$\Phi(F_{\alpha_{i_1}} \cdots F_{\alpha_{i_k}} v_\lambda) = F_{\alpha_{i_1}} \cdots F_{\alpha_{i_k}} w_\lambda.$$

The point is that all relations among such vectors are determined by the Lie algebra relations and by the highest weight λ . Indeed, the action of the Cartan subalgebra is fixed by

$$Hv_\lambda = \lambda(H)v_\lambda,$$

and the positive root operators kill the highest weight vector:

$$E^{\alpha_i} v_\lambda = 0.$$

The remaining commutation relations are the same in both modules, because they come from the Lie algebra $\mathfrak{g}$ itself. Hence any linear relation among lowering-operator monomials applied to v_λ also holds for w_λ . Therefore Φ is well-defined.

By construction, Φ commutes with the action of the lowering operators, the Cartan subalgebra, and the raising operators. Hence Φ is a $\mathfrak{g}$ -module homomorphism.

Moreover, $\Phi(v_\lambda) = w_\lambda \neq 0$, so Φ is not the zero map. Since V and W are irreducible, the kernel of Φ is either 0 or all of V . It cannot be all of V , because $\Phi(v_\lambda) \neq 0$. Thus

$$\ker \Phi = 0.$$

Similarly, the image of Φ is a nonzero submodule of W , hence by irreducibility

$$\operatorname{im} \Phi = W.$$

Therefore Φ is an isomorphism.

Thus two irreducible highest weight modules with the same highest weight are isomorphic.

Exercise 2. Representations and tensor decompositions for type $B_2 \cong C_2$

Throughout, $V = \mathbb{C}^4$ is the defining representation of $\mathfrak{sp}(4)$, equipped with a nondegenerate invariant symplectic form

$$\Omega \in \Lambda^2 V^*.$$

The dimensions of the important irreducibles are denoted by

$$\mathbf{4}, \quad \mathbf{5}, \quad \mathbf{10}, \quad \mathbf{1}.$$

Part I. Decomposing $\mathbf{4} \otimes \mathbf{4}$ from the symplectic and volume forms

(a) The decomposition of $\Lambda^2 V$

Solution. The vector space $\Lambda^2 V$ has dimension

$$\dim \Lambda^2 V = \binom{4}{2} = 6.$$

The symplectic form $\Omega \in \Lambda^2 V^*$ defines a contraction map

$$c : \Lambda^2 V \longrightarrow \mathbb{C}.$$

Since Ω is nonzero and nondegenerate, this contraction is not the zero map. Choose a bivector $\omega \in \Lambda^2 V$ such that

$$c(\omega) \neq 0.$$

Then $\mathbb{C}\omega$ is a one-dimensional complement to the kernel of c .

Define

$$W := \ker c.$$

Since c is a nonzero linear map from a 6-dimensional vector space to a 1-dimensional vector space, its kernel has dimension

$$\dim W = 5.$$

Therefore

$$\Lambda^2 V = \mathbb{C}\omega \oplus W.$$

This decomposition is invariant under $\mathfrak{sp}(4)$. Indeed, Ω is invariant by definition of $\mathfrak{sp}(4)$, so the contraction map c is an intertwiner. Hence its kernel is an invariant subspace. The line $\mathbb{C}\omega$ is the trivial one-dimensional representation, while W is the complementary 5-dimensional representation.

Thus

$$\Lambda^2 V \cong \mathbf{5} \oplus \mathbf{1}.$$

(b) The invariant quadratic form on W

Solution. The wedge product gives a bilinear map

$$\Lambda^2 V \times \Lambda^2 V \longrightarrow \Lambda^4 V.$$

Since $\Lambda^4 V$ is one-dimensional, we identify it with $\mathbb{C}$ using the invariant volume form

$$\text{vol} = \Omega \wedge \Omega.$$

This gives a bilinear form

$$q : \Lambda^2 V \times \Lambda^2 V \longrightarrow \mathbb{C}.$$

For two-forms, wedge product is symmetric because

$$\alpha \wedge \beta = (-1)^{2 \cdot 2} \beta \wedge \alpha = \beta \wedge \alpha.$$

Thus q is symmetric.

The form q is invariant because the wedge product and the volume form are invariant under $\mathfrak{sp}(4)$. Its restriction to the one-dimensional line $\mathbb{C}\omega$ is nonzero, and the decomposition

$$\Lambda^2 V = \mathbb{C}\omega \oplus W$$

can be chosen so that W is the orthogonal complement of this line. Hence $q|_W$ is nondegenerate.

Therefore $\mathfrak{sp}(4)$ acts on the 5-dimensional vector space W preserving a nondegenerate symmetric bilinear form. This gives a Lie algebra homomorphism

$$\rho : \mathfrak{sp}(4) \longrightarrow \mathfrak{so}(W) \cong \mathfrak{so}(5).$$

(c) Showing $\mathfrak{sp}(4) \cong \mathfrak{so}(5)$

Solution. We have constructed a homomorphism

$$\rho : \mathfrak{sp}(4) \longrightarrow \mathfrak{so}(5).$$

The Lie algebra $\mathfrak{sp}(4)$ is simple, so the kernel of ρ is an ideal. Hence

$$\ker \rho = 0 \quad \text{or} \quad \ker \rho = \mathfrak{sp}(4).$$

The action on W is not trivial, so $\rho \neq 0$. Therefore

$$\ker \rho = 0,$$

and ρ is injective.

Now compare dimensions. We have

$$\dim \mathfrak{sp}(2n) = n(2n + 1).$$

For $n = 2$, this gives

$$\dim \mathfrak{sp}(4) = 2(5) = 10.$$

Also

$$\dim \mathfrak{so}(5) = \frac{5 \cdot 4}{2} = 10.$$

So ρ is an injective linear map between Lie algebras of the same dimension. Hence it is an isomorphism:

$$\boxed{\mathfrak{sp}(4) \cong \mathfrak{so}(5).}$$

(d) The symmetric square $\text{Sym}^2 V$

Solution. The tensor square decomposes as

$$V \otimes V = \text{Sym}^2 V \oplus \Lambda^2 V.$$

Since $\dim V = 4$, we have

$$\dim \text{Sym}^2 V = \frac{4 \cdot 5}{2} = 10.$$

The representation V has highest weight

$$\omega_2 = \frac{1}{2}(L_1 + L_2)$$

according to the convention in the sheet. Therefore the highest weight of $\text{Sym}^2 V$ is twice this:

$$2\omega_2 = L_1 + L_2.$$

This is the highest root of the C_2 root system. The irreducible representation with highest weight equal to the highest root is the adjoint representation. Hence

$$\text{Sym}^2 V \cong \mathfrak{sp}(4).$$

Since $\dim \text{Sym}^2 V = 10 = \dim \mathfrak{sp}(4)$, we conclude

$$\boxed{\text{Sym}^2 V \cong \mathbf{10}.}$$

(e) Assembling the decomposition

Solution. Combining the symmetric and antisymmetric decompositions, we obtain

$$\mathbf{4} \otimes \mathbf{4} = \text{Sym}^2 \mathbf{4} \oplus \Lambda^2 \mathbf{4}.$$

From part (d),

$$\text{Sym}^2 \mathbf{4} \cong \mathbf{10}.$$

From part (a),

$$\Lambda^2 \mathbf{4} \cong \mathbf{5} \oplus \mathbf{1}.$$

Therefore

$$\boxed{\mathbf{4} \otimes \mathbf{4} \cong \mathbf{10} \oplus \mathbf{5} \oplus \mathbf{1}.}$$

Equivalently,

$$\boxed{\text{Sym}^2 V \cong \mathbf{10}, \quad \Lambda^2 V \cong \mathbf{5} \oplus \mathbf{1}.}$$

The singlet is the invariant bivector dual to the symplectic form. Concretely, if Ω is used to identify V with V^* , then the inverse bivector

$$\omega = \Omega^{-1} \in \Lambda^2 V$$

spans the trivial representation.

Part II. Weight diagrams

(a) The B_2 root system

Solution. In the (L_1, L_2) -plane, the B_2 root system consists of

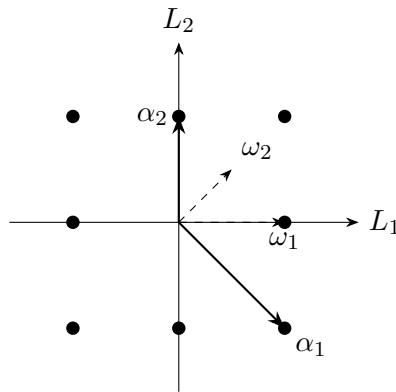
$$\pm L_1, \quad \pm L_2, \quad \pm L_1 \pm L_2.$$

The simple roots are

$$\alpha_1 = L_1 - L_2, \quad \alpha_2 = L_2.$$

The fundamental weights are

$$\omega_1 = L_1, \quad \omega_2 = \frac{1}{2}(L_1 + L_2).$$



(b) Weight diagrams of 4 and 5

Solution. The defining representation **4** of $\mathfrak{sp}(4)$ has weights

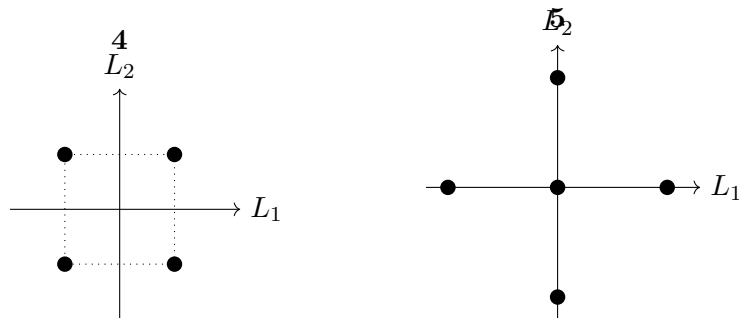
$$\frac{1}{2}(L_1 + L_2), \quad \frac{1}{2}(L_1 - L_2), \quad \frac{1}{2}(-L_1 + L_2), \quad -\frac{1}{2}(L_1 + L_2).$$

Thus its weight diagram is a square centered at the origin.

The representation **5** is the vector representation of $\mathfrak{so}(5)$ under the isomorphism $\mathfrak{sp}(4) \cong \mathfrak{so}(5)$. Its weights are

$$L_1, -L_1, L_2, -L_2, 0.$$

So it is a cross with one weight at the origin.



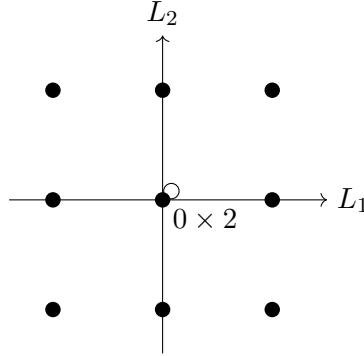
(c) Weight diagram of the adjoint $\mathbf{10}$

Solution. The adjoint representation has weights equal to the roots, together with the zero weight of multiplicity equal to the rank. Since $\mathfrak{sp}(4)$ has rank 2, the zero weight has multiplicity 2.

Thus the weights of $\mathbf{10}$ are

$$\pm L_1, \quad \pm L_2, \quad \pm L_1 \pm L_2,$$

plus two copies of 0.



(d) Overlaying the weights of $\mathbf{4} \otimes \mathbf{4}$

Solution. The weights of a tensor product are obtained by adding weights. Therefore the weights of $\mathbf{4} \otimes \mathbf{4}$ are all sums

$$\mu + \nu, \quad \mu, \nu \in \text{Wt}(\mathbf{4}).$$

Since $\mathbf{4}$ has four weights, the tensor product has 16 weights counted with multiplicity.

Graphically, these weights split into three pieces:

$$\mathbf{4} \otimes \mathbf{4} \cong \mathbf{10} \oplus \mathbf{5} \oplus \mathbf{1}.$$

The $\mathbf{10}$ contributes the eight nonzero roots and two copies of zero. The $\mathbf{5}$ contributes

$$\pm L_1, \quad \pm L_2, \quad 0,$$

and the singlet contributes one further zero weight.

Thus the full tensor product has the following multiplicities:

weight	multiplicity in $\mathbf{4} \otimes \mathbf{4}$
$\pm(L_1 + L_2), \pm(L_1 - L_2)$	1
$\pm L_1, \pm L_2$	2
0	4

These multiplicities match

$$\mathbf{10} \oplus \mathbf{5} \oplus \mathbf{1}.$$

(e) Decomposing $\mathbf{4} \otimes \mathbf{5}$

Solution. We use highest weights and dimensions. The highest weight of $\mathbf{4}$ is

$$\omega_2 = \frac{1}{2}(L_1 + L_2),$$

and the highest weight of **5** is

$$\omega_1 = L_1.$$

Therefore the largest possible highest weight in $\mathbf{4} \otimes \mathbf{5}$ is

$$\omega_1 + \omega_2 = L_1 + \frac{1}{2}(L_1 + L_2) = \frac{3}{2}L_1 + \frac{1}{2}L_2.$$

The irreducible representation with this highest weight has dimension 16.

Since

$$\dim(\mathbf{4} \otimes \mathbf{5}) = 4 \cdot 5 = 20,$$

there must be another summand of dimension 4. The natural remaining summand is the defining representation **4**.

Hence

$$\boxed{\mathbf{4} \otimes \mathbf{5} \cong \mathbf{16} \oplus \mathbf{4}.}$$

This is the B_2/C_2 analogue of the familiar rule that tensoring by a small fundamental representation produces a highest-weight summand plus lower summands obtained by subtracting roots.

Exercise 3. Adding matter to Yang-Mills theory

Let (ρ, V) be a finite-dimensional representation of a Lie group G , and let

$$\rho_* : \mathfrak{g} \longrightarrow \mathfrak{gl}(V)$$

be the induced Lie algebra representation. A matter field is a smooth map

$$\phi : \mathbb{R}^4 \longrightarrow V.$$

Under a gauge transformation $g : \mathbb{R}^4 \rightarrow G$, it transforms as

$$\phi \longmapsto \phi^g = \rho(g)\phi.$$

The gauge-covariant derivative is

$$D_i \phi = \partial_i \phi + \rho_*(A_i)\phi.$$

(a) Transformation law of the covariant derivative

Solution. The gauge field transforms as

$$A_i^g = \text{Ad}_g(A_i) - (\partial_i g)g^{-1},$$

with the sign convention compatible with

$$D_i = \partial_i + \rho_*(A_i).$$

We want to show that

$$(D_i \phi)^g = \rho(g)D_i \phi.$$

Compute the transformed covariant derivative:

$$D_i^g \phi^g = \partial_i(\rho(g)\phi) + \rho_*(A_i^g)\rho(g)\phi.$$

The first term is

$$\partial_i(\rho(g)\phi) = (\partial_i \rho(g))\phi + \rho(g)\partial_i \phi.$$

For the second term, use

$$\rho_*(\text{Ad}_g X) = \rho(g)\rho_*(X)\rho(g)^{-1}$$

and

$$\rho_*((\partial_i g)g^{-1}) = (\partial_i \rho(g))\rho(g)^{-1}.$$

Thus

$$\begin{aligned}\rho_*(A_i^g)\rho(g)\phi &= \rho_*(\text{Ad}_g A_i)\rho(g)\phi - \rho_*((\partial_i g)g^{-1})\rho(g)\phi \\ &= \rho(g)\rho_*(A_i)\phi - (\partial_i \rho(g))\phi.\end{aligned}$$

Adding both contributions gives

$$\begin{aligned}D_i^g \phi^g &= (\partial_i \rho(g))\phi + \rho(g)\partial_i \phi + \rho(g)\rho_*(A_i)\phi - (\partial_i \rho(g))\phi \\ &= \rho(g)(\partial_i \phi + \rho_*(A_i)\phi) \\ &= \rho(g)D_i \phi.\end{aligned}$$

Therefore

$$\boxed{D_i^g \phi^g = \rho(g)D_i \phi.}$$

This means that $D_i \phi$ transforms in the same representation as ϕ itself.

(b) Gauge invariance of the matter action

Solution. Assume V carries a G -invariant Hermitian inner product

$$\langle \cdot, \cdot \rangle.$$

This means

$$\langle \rho(g)v, \rho(g)w \rangle = \langle v, w \rangle$$

for all $g \in G$ and $v, w \in V$.

From part (a),

$$D_i \phi \mapsto \rho(g)D_i \phi.$$

Therefore

$$\begin{aligned}\langle D_i^g \phi^g, D_i^g \phi^g \rangle &= \langle \rho(g)D_i \phi, \rho(g)D_i \phi \rangle \\ &= \langle D_i \phi, D_i \phi \rangle.\end{aligned}$$

Similarly,

$$\langle \phi^g, \phi^g \rangle = \langle \rho(g)\phi, \rho(g)\phi \rangle = \langle \phi, \phi \rangle.$$

Hence the integrand

$$\sum_i \langle D_i \phi, D_i \phi \rangle + m^2 \langle \phi, \phi \rangle$$

is gauge invariant pointwise. Therefore the matter action

$$S_{\text{mat}}[A, \phi] = \int_{\mathbb{R}^4} \left(\sum_i \langle D_i \phi, D_i \phi \rangle + m^2 \langle \phi, \phi \rangle \right) d^4 x$$

is gauge invariant.

Since the Yang-Mills action S_{YM} is also gauge invariant, the total action

$$S_{\text{YM}} + S_{\text{mat}}$$

is gauge invariant.

(c) **The case** $G = SU(2)$ **and** $V = \mathbb{C}^2$

Solution. Let $V = \mathbb{C}^2$ be the fundamental representation of $SU(2)$. Write the scalar doublet as

$$\phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}.$$

Choose a basis T_a of $\mathfrak{su}(2)$, for example

$$T_a = \frac{i}{2} \sigma_a,$$

where σ_a are the Pauli matrices. Write the gauge field as

$$A_i = A_i^a T_a.$$

Then the covariant derivative is

$$D_i \phi = \partial_i \phi + A_i^a T_a \phi.$$

In components,

$$(D_i \phi)^\alpha = \partial_i \phi^\alpha + A_i^a (T_a)^\alpha_\beta \phi^\beta, \quad \alpha, \beta = 1, 2.$$

Using the standard Hermitian inner product on $\mathbb{C}^2$,

$$\langle u, v \rangle = u^\dagger v,$$

we get

$$\langle D_i \phi, D_i \phi \rangle = (D_i \phi)^\dagger (D_i \phi) = \sum_{\alpha=1}^2 \overline{(D_i \phi)^\alpha} (D_i \phi)^\alpha.$$

Thus the matter kinetic term is

$$\sum_i (D_i \phi)^\dagger (D_i \phi).$$

Now suppose instead that

$$V = \mathfrak{g}, \quad \rho = \text{Ad}.$$

Then a matter field is a Lie-algebra-valued field

$$\phi : \mathbb{R}^4 \rightarrow \mathfrak{g},$$

and the induced representation is

$$\rho_*(X)Y = [X, Y].$$

Hence the covariant derivative becomes

$$D_i \phi = \partial_i \phi + [A_i, \phi].$$

This is exactly the covariant derivative in the adjoint representation. It is the same form used for the curvature and for adjoint-valued scalar fields.